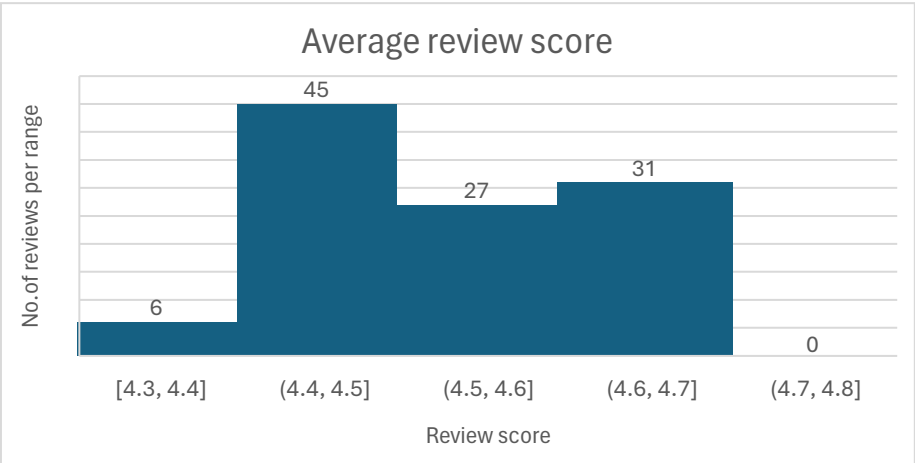
Hotel Revenue Statistical Analysis

Descriptive Statistics

	France	Germany	Italy	Spain	Uk	USA
Mean	82230.76923	89142.85714	87416.66667	88538.46154	84428.57143	80758.62069
Standard Error	726.3700243	602.714946	924.8942063	1375.683441	480.9288066	529.0924761
Median	82000	90000	88000	91000	85000	80000
Mode	82000	90000	88000	91000	85000	80000
Standard Deviation	2618.964368	2761.986862	3203.927514	4960.097187	2203.89266	2849.250182
Sample Variance	6858974.359	7628571.429	10265151.52	24602564.1	4857142.857	8118226.601
Kurtosis	0.98429176	6.736401349	1.540597323	1.851653416	6.418238936	11.43785518
Skewness	0.361068093	-2.776136404	-1.429656753	-1.727032967	-2.620258194	3.535689109
Range	9000	10000	10000	15000	9000	12000
Minimum	78000	81000	81000	78000	78000	79000
Maximum	87000	91000	91000	93000	87000	91000
Sum	1069000	1872000	1049000	1151000	1773000	2342000
Count	13	21	12	13	21	29

Using important statistical metrics, Table analyses the revenue from six visitor countries (France, Germany, Italy, Spain, the UK, and the USA). In order to spot trends and variations in revenue, it highlights key measures like mean, median, mode, standard deviation, range, skewness and kurtosis. For instance, the USA has the highest total revenue(2342000) and count(29) Germany has the highest average revenue(89142.8). Spain, with a range of 15000 and a standard deviation of 4960, shows the highest variability, indicating diverse revenue performance. Asymmetries or outliers are suggested by skewness and kurtosis, which show the forms of data distributions. For example, Germany's left skewness(-2.77) and the USA's right skewness(3.53). This analysis aids in comprehending the performance of each country, identifying stable markets with low variability, such as the UK, and focusing on high-revenue markets, such as Spain and the USA.

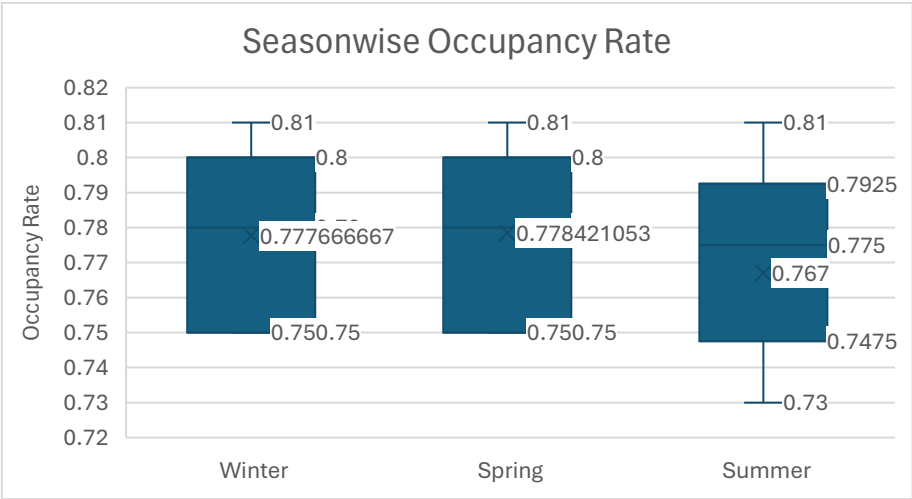
Histogram:



The histogram displays the distribution of average review score intervals. Most reviews fall in the range of (4.4, 4.5] (45), indicating a strong positive perception, followed by significant frequencies in (4.5, 4.6] (27) and (4.6, 4.7] (31), confirming overall satisfaction. Less extreme ratings are shown by the small number of reviews in the [4.3,4.4] (6), and no reviews are in the

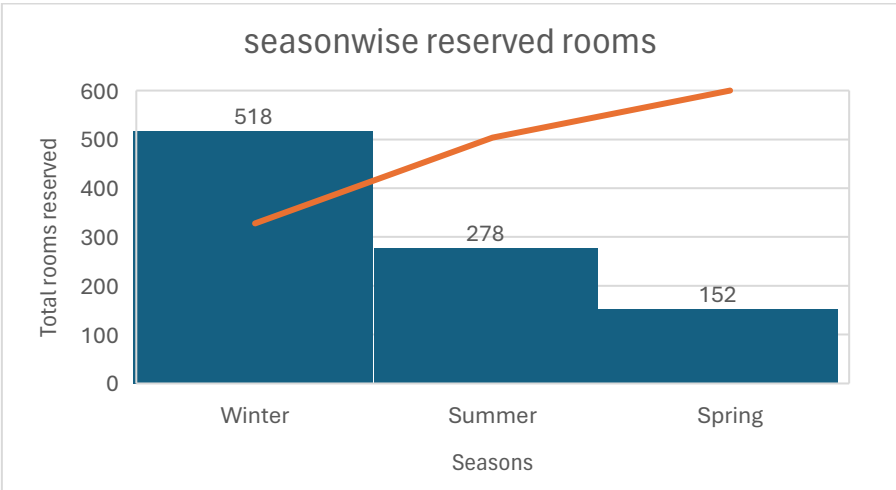
range of (4.7, 4.8] and over. This analysis highlights the central tendency of customer reviews, helping identify satisfaction trends and areas that require development. The visualization effectively summarizes customer feedback, providing actionable insights to maintain high performance and enhance weaker areas.

Box and Whisker Plot:



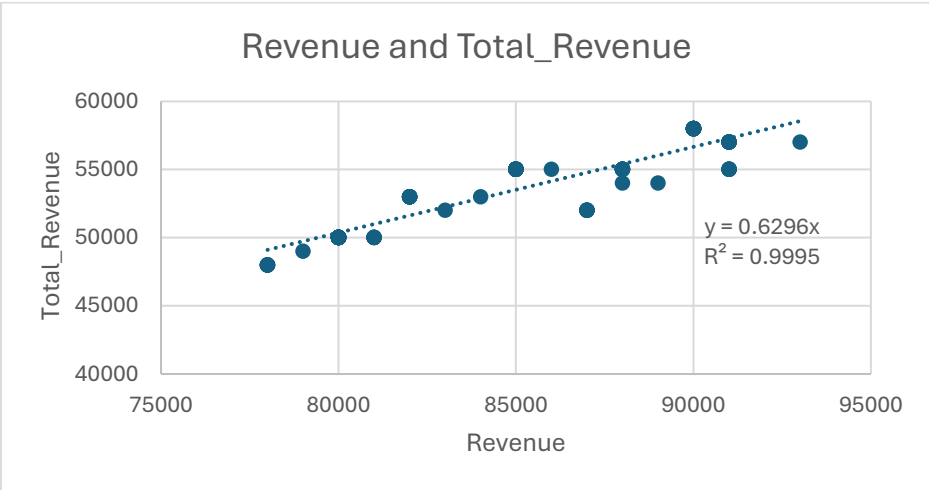
A comparison of winter, spring and summer occupancy rates is shown in the Box and Whisker plot. The median occupancy rate for the winter and spring seasons is 0.78, suggesting that central trends are consistent over these times, but both have a different mean value and similar whisker range (0.75 to 0.81) and a comparatively small interquartile range (IQR), with values between 0.75 and 0.8, indicating stable occupancy levels. However, summer, on the other hand, exhibits a somewhat lower median (0.775), mean (0.767) and broader IQR from 0.7475 to 0.7925, indicating greater variability in occupancy. Notably, summer shows the lowest minimum (0.73) and the largest range overall, indicating variations in occupancy. Here, the box and whisker plot is especially helpful because it shows the variability, spread and central tendency of occupancy rates across all seasons.

Pareto Chart:



A Pareto chart, which combines a cumulative percentage line and a bar chart, is a useful tool for determining the most important components in a dataset. Winter has the most reservations(518), followed by summer(278) and spring(152). The cumulative percentage line shows that more than 80% of reservations are made during the winter and summer, while 100% of reservations are made throughout all three seasons combined. This figure is very helpful for setting priorities because it illustrates how concentrating on the winter and summer months could have the biggest impact. By applying the Pareto principle, decision-makers can allocate resources and strategies more effectively to seasons driving the most bookings.

Scatter Plot:



The relationship between total revenue and revenue is visualised via the scatter plot. In the plot, Individual data points are represented by the points showing the total revenue and revenue for a specific observation. The trend line represented by the equation y=0.6296x indicates the overall trend in the data. The two variables have a strong positive correlation, as indicated by the R-squared value of 0.9995, which means that revenue increases total revenue tends to increase as well. In order to better comprehend the interactions between the variables, scatter plots are useful for spotting patterns and trends in data.

Regression Statistics

Regression Statistics	
Multiple R	0.987278174
R Square	0.974718193
Adjusted R Square	0.974481914
Standard Error	719.0536934
Observations	109

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	2132933792	2132933792	4125.292	2.76646E-87
Residual	107	55323088.89	517038.214		
Total	108	2188256881			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	51998.05485	517.1425809	100.5487786	9.2E-108	50972.88	53023.22971	50972.88	53023.22971
Marketing_Spend	0.549076055	0.0085488	64.22843884	2.77E-87	0.532129057	0.566023053	0.532129057	0.566023053

Multiple R: The value is ≈0.988 is very near to 1. This indicates that the independent variable and the dependent variable have a strong positive correlation. In simpler terms, revenue typically increases when marketing spending increases.

R Square: The value is ≈0.974 or ≈97.47%. In this variation, the dependent variable is explained by the independent variable, which means ≈97.47% of the variation in revenue can be explained by the variation in marketing spend. This says that marketing spend is a significant predictor of revenue.

Adjusted R Square: The value is ≈0.974 is extremely near to the R Square, which implies that the model is not overfitting the data.

Standard Error: The Standard Error is 719.0536934. This figure indicates the average distance of the data points from the regression line. A smaller value suggests that the model fits the data more accurately.

Analysis of Variance

Regression: This row shows the variance that the regression model considers. The regression model is clearly statistically significant, as evidenced by the high F-statistic (4125.292) and the extremely low significance F value (2.76646E-87).

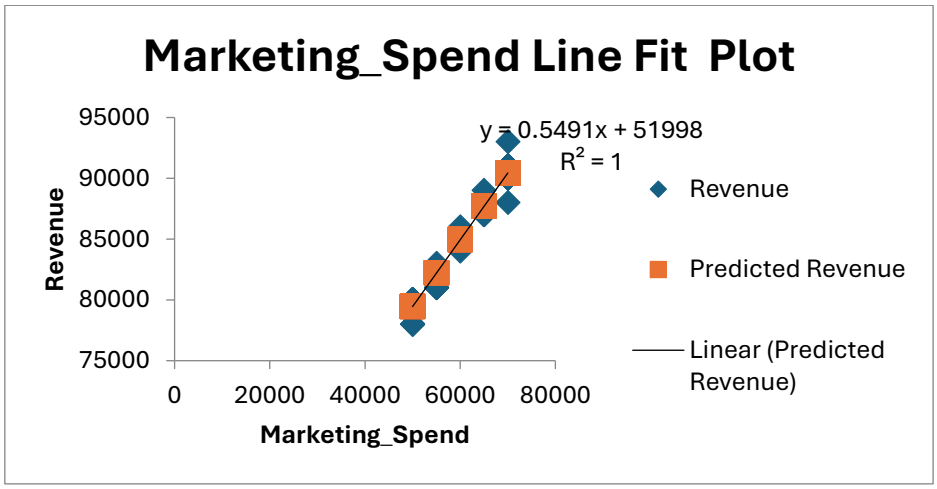
Residual: This row displays the model's error or the variation that cannot be explained. The model explains a significant amount of the data variability, as shown by the residual sum of squares (SS), which is comparatively small compared to the regression SS.

Coefficients

Intercept: The intercept is 51,998.05, representing the projected revenue when marketing expenses are nil. This shows a base revenue level even without any marketing investment.

Marketing spend coefficient: the marketing spend coefficient is 0.549, which means that every additional unit spent on marketing leads to a revenue increase of about 0.549 units. This coefficient indicates a positive link between marketing spending and revenue, demonstrating that marketing efforts have a direct and measurable impact on revenue generation.

Line Fit Plot:



The Line Fit Plot demonstrates a strong positive correlation between marketing spend and revenue. The orange line shows the regression model's predicted values and the blue dots show the actual data points. With an equation of y=0.5491x+51998 and an $R2$ value of 1, the regression line displays an upward trend, suggesting that the model accounts for all revenue variability. Revenue rises by roughly 0.549 units for every unit increase in marketing spend.

This analysis is useful for firms when optimising marketing budgets since it provides exact insights into the influence of marketing spending on revenue growth. However, because the model assumes a linear relationship and excludes other potential revenue-influencing factors, its usefulness may be limited.

Further analysis, such as residual evaluation, multicollinearity checks, and sensitivity testing, can help confirm the results and provide more trustworthy decision-making for future marketing strategies.